

High-Energy Laboratory Astrophysics with Accelerator Beams

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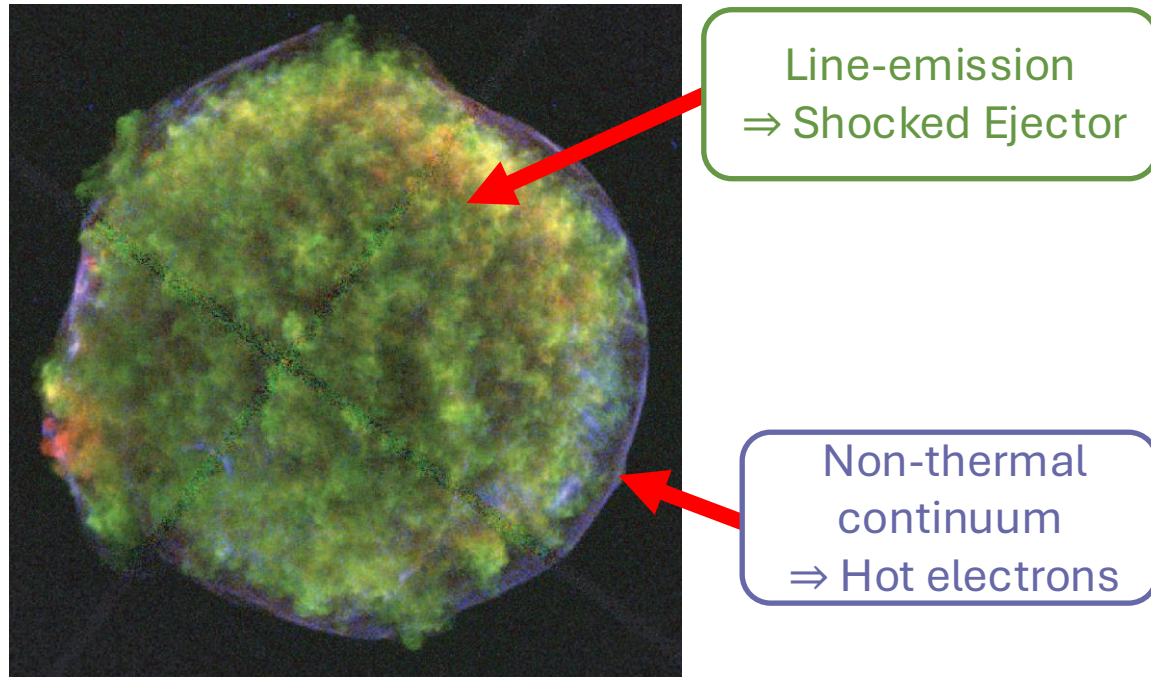
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Overview

- **Astrophysical problem** – How do collisionless plasmas exchange energy between bulk flows, electromagnetic fluctuations, and non-thermal particles?
- **Laboratory strategy:** use accelerator beams as controllable fast-particle populations
- **Case study 1:** Ion acceleration / diffusion by laser driven, magnetized plasma:
 - Experimental design
 - Time of Flight / interferometry
- **Case study 2:** Relativistic pair jets interacting with an ambient RF plasma:
 - Experimental design
 - Faraday rotation diagnostic
 - Application to Blazar jets
- **Outlook:** next-generation laser facilities can turn these into systematic, higher-repetition, diagnostic-rich studies.

Extreme astrophysical environments are governed by collisionless plasma processes



Tycho's SNR: non-thermal X-ray rims trace particle acceleration at the blast wave; line emission maps shocked ejecta.

J. Warren et al, ApJ **634**, 376 (2005).

- Blazars, pulsar winds and supernova remnants generate cosmic rays and non-thermal radiation.
- Key questions in the microphysics: particle energisation, magnetic-field generation and transport
- Tycho shows the problem clearly: X-rays reveal efficient acceleration, but not the microscopic process behind it
- Beam–plasma experiments provide controlled tests of these processes.

Particle accelerators generate fast-particle populations with well-controlled, reproducible characteristics

- **Primary beam species:** electrons, protons or ions, with selected charge state and bunch structure.
- **Known input distribution:** MeV–GeV energies, μm –cm spot sizes and mrad-scale divergence, depending on species and beamline.
- **Timing:** beam/plasma overlap can be synchronized on ns– μs scales.
- **Secondary targets:** generate photons, neutrons, positrons, fragments or broadband spectra from a controlled primary beam.



Case study 1
UNILAC → Z6, GSI
 $^{52}\text{Cr}^{N+}$, 8.7 MeV/u



Case study 2
SPS → HiRadMat, CERN
440 GeV protons,

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Measurement of ion acceleration and diffusion in a laser-driven magnetized plasma

Josh T. Chu¹, **Jack. W. D. Halliday**², C. Heaton¹, Kassie Moczulski³, A. Blazevic⁴, M. Metternich^{4,5}, H. Nazary^{4,5}, D. Schumacher⁴, H. Wen³, T. Campbell¹, A. Scopatz³, C. A. J. Palmer⁶, A. F. A. Bott¹, C.D. Arrowsmith¹, C. Stuart¹, V. Bagnoud^{4,5}, S. Feister⁷, O.Karnbach¹, P. Neumayer⁴, A. Reyes³, E. Hansen³, C. Spindloe², S. Sakar¹, A. R. Bell², F. Miniati¹, A. A. Schekochihin¹, B. Reville⁹, D. Q. Lamb¹⁰, P. Tzeferacos³, R.Bingham^{2,8} and Gianluca. Gregori¹

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Measurement of ion acceleration and diffusion in a laser-driven magnetized plasma

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Physics of Plasmas

ARTICLE

pubs.aip.org/aip/pop

Numerical simulations of laser-driven experiments of ion acceleration in stochastic magnetic fields

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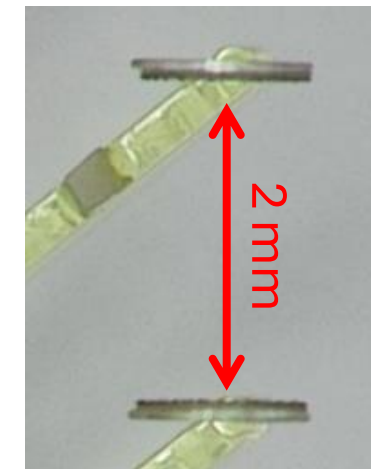
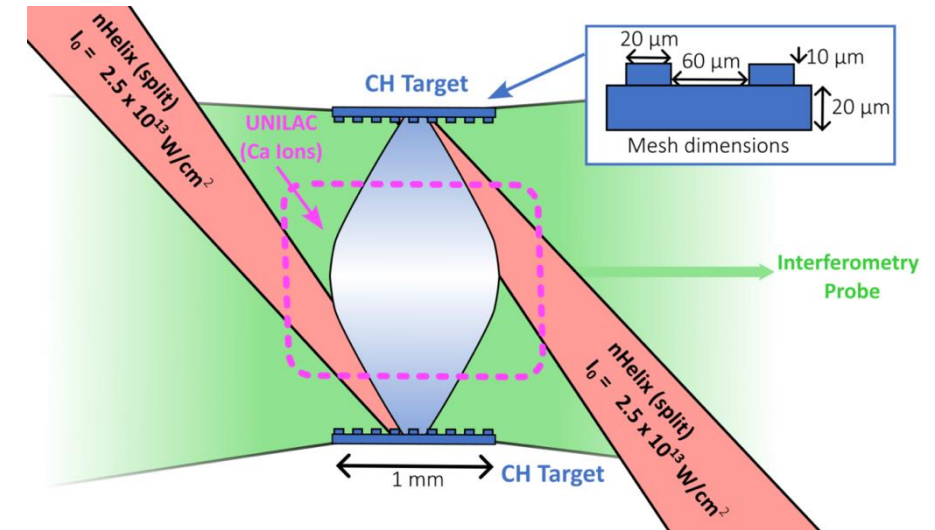
K. Moczulski^{1,a}, H. Wen¹, T. Campbell², A. Scopatz¹, C. A. J. Palmer³, A. F. A. Bott², C. D. Arrowsmith², K. A. Beyer⁴, A. Blazevic⁵, V. Bagnoud⁵, S. Feister⁶, J. Halliday^{2,7}, O. Karnbach², M. Metternich⁵, H. Nazary⁵, P. Neumayer⁵, A. Reyes⁶, E. C. Hansen⁶, D. Schumacher⁵, C. Spindloe², S. Sakar⁴, A. R. Bell⁴, R. Bingham^{2,8}, F. Miniati⁴, A. A. Schekochihin⁴, B. Reville⁹, D. Q. Lamb¹⁰, G. Gregori¹ and P. Tzeferacos¹

By combining high-power lasers with the GSI UNILAC, we can study ion acceleration in a laser-driven plasma

- Plastic (CH) targets with pre-imposed 60 μm spatial modulations are driven by nHelix laser beams ($2 \times 30 \text{ J}$ in **10 ns**) to generate counter-propagating plasma outflows.
- The resulting plasmas develop strong density and temperature gradients – but **no evidence** of strong fluid-scale turbulence is observed
- UNILAC-accelerated ions are sent through this plasma, and the change in their energy is measured via time-of-flight diagnostics.

^{52}Cr ions: 14+, 8.75 MeV/u or 20+, 8.66 MeV/u

- Despite the apparent absence of fluid turbulence, ion acceleration and diffusion are observed → **kinetic scale instability**



The goal of this experiment is to benchmark models of cosmic-ray acceleration in the laboratory

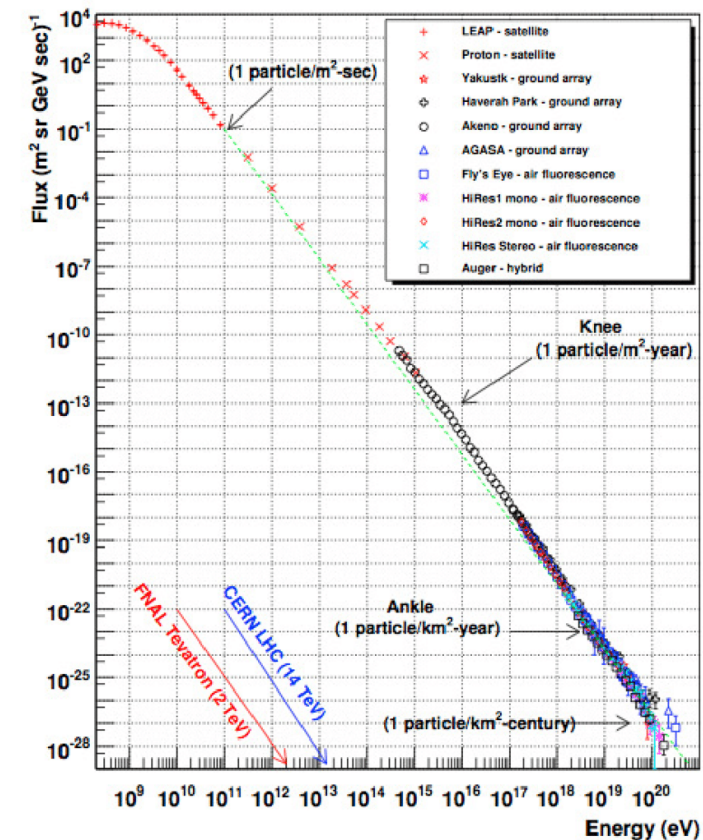
- **Cosmic rays reach up to 10^{20} eV**, but the mechanisms and environments responsible for such extreme energies remain debated

- **2nd order Fermi acceleration** – The leading large-scale model, where particles gain energy through stochastic scattering from moving magnetic structures:

$$\Delta E \propto Z u_{\text{turb}} B \sqrt{L_{\text{sys}} l_B}$$

- **Kinetic plasma turbulence** – At smaller scales, plasma turbulence can transfer energy directly to ions through coherent wave–particle coupling. For Lower hybrid waves, scales as:

$$\Delta E \propto Z^2 \sqrt{V_b L}$$



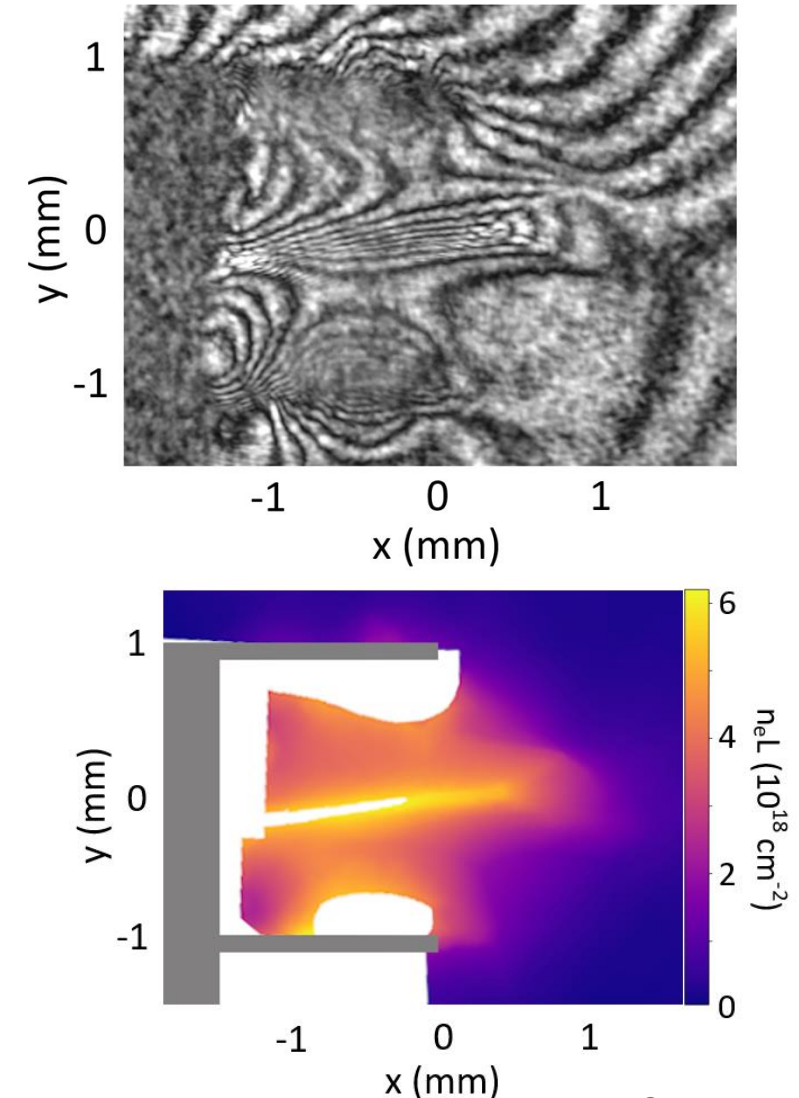
A. López-Oramas “Multi-year campaign of the gamma-ray binary LS I +61° 303 and search for VHE emission from gamma-ray binary candidates with the MAGIC telescopes” PhD thesis. 2015

Interferometry shows a smooth inflow without obvious signatures of fluid-scale turbulence

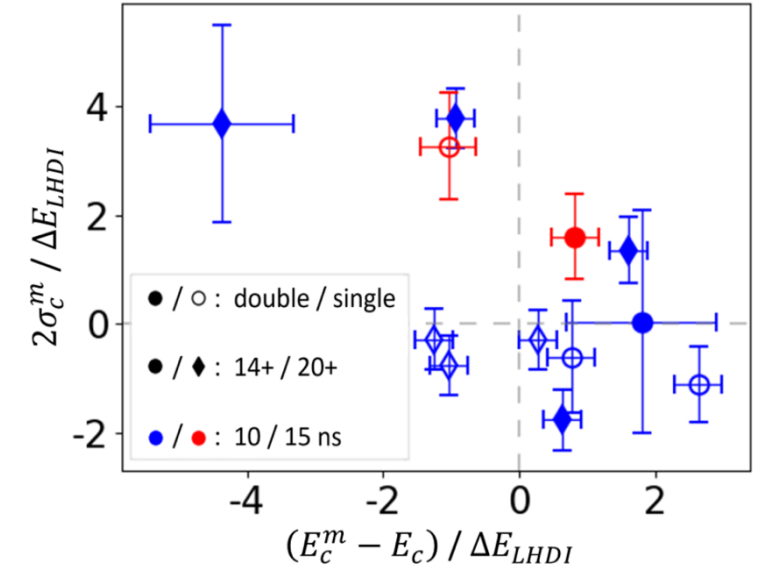
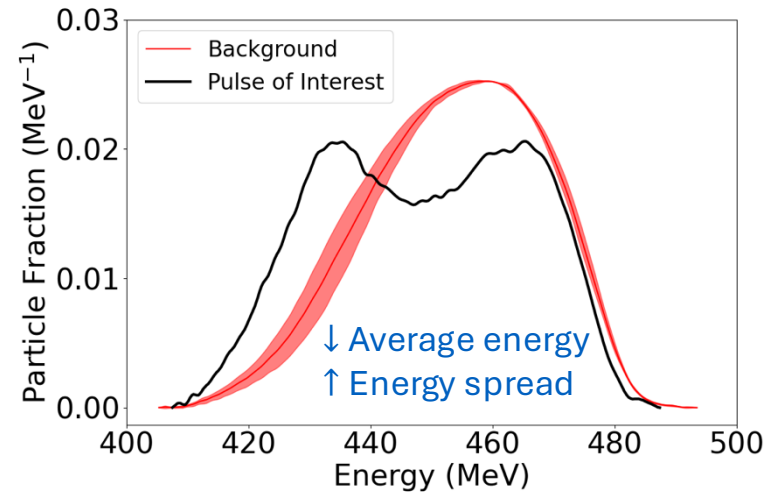
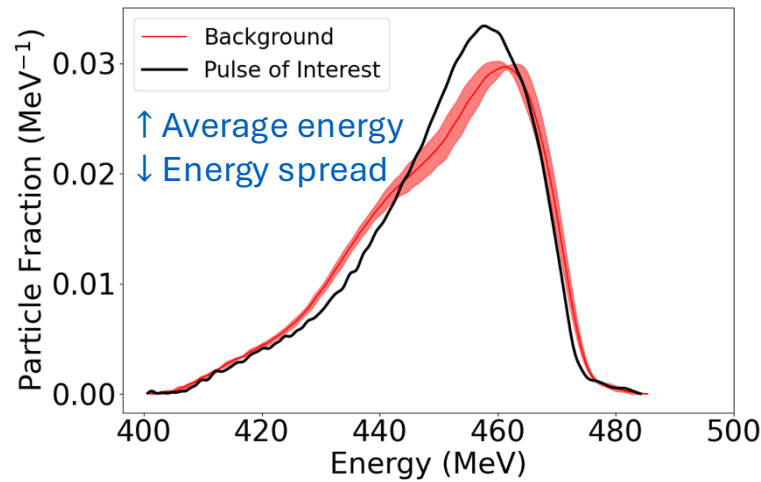
- Fringe shifts in interferograms are proportional to the line-integrated electron density of the plasma
- Qualitatively, the interaction region is smooth and **lacks stochastic small-scale structure** expected from fluid turbulence
- Quantitative electron density maps ($n_e \sim 5 \times 10^{18}$) were obtained via manual fringe tracing [G. F. Swadling *et. al.* ; J. D. Hare *et. al.*]
- These measurements provide **bounds** on both the plasma temperature and the RMS velocity-fluctuation level associated with fluid turbulence (more later)

G. F. Swadling *et. al.* “Oblique shock structures formed during the ablation phase of aluminium wire array z-pinches” *Physics of Plasmas*, 2013.

J. D. Hare *et. al.* “Two-colour interferometry and Thomson scattering measurements of a plasma gun” *Plasma Physics and Controlled Fusion*, 2019



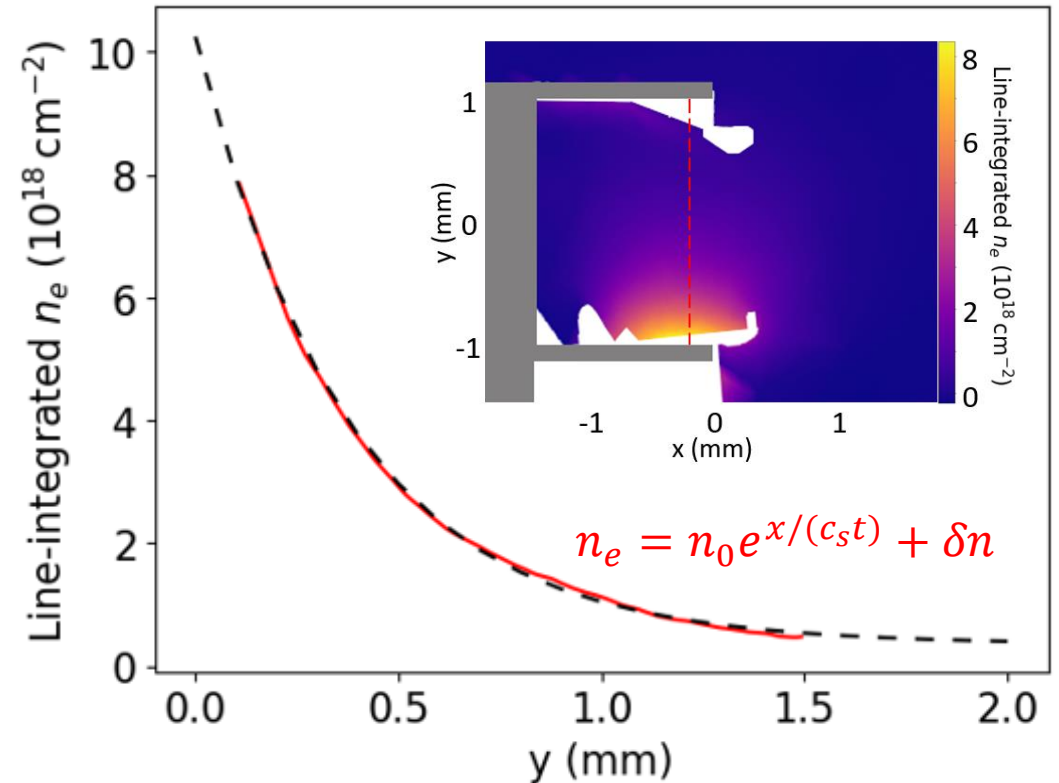
Energy shifts and broadening appear in the spectra of ions that are propagated through the plasma



- Cadmium ions are propagated through the interaction region, and their spectra are measured using time-of-flight
- Although the spectra vary shot-to-shot, ion acceleration or diffusion is observed on every shot
- The scatter plot is normalised to the expected energy shift from the lower-hybrid (kinetic) instability for the experimental parameters: $\Delta E_{LHDI} = 50 \bar{Z}_i$ [MeV], $E_c = 450$ MeV. Only $\bar{Z}_i$ varies between shots.
- Expected broadening from Coulomb collisions is negligible (≈ 1 eV)

Isothermal expansion of the inflows set a constraint on the electron temperature

- Density profiles from single-target shots were fit assuming an isothermal-expansion
- The fitted scale height yields an inflow sound speed $\rightarrow$ inflow electron temperature ($T_{\text{in}} \sim 120$ eV), following the approach of D. Yuan *et al.*
- This is a coarse estimate which should be interpreted as a lower bound on the temperature in the interaction region
- To invalidate the claim that fluid turbulence is negligible, the required peak temperature in the interaction region would exceed ~ 750 keV – which is non-physical



D. Yuan et al. “*Electron stochastic acceleration in laboratory-produced kinetic turbulent plasmas*” Nat. Comms. (2024).

Interferometry provides a quantitative bound on the RMS velocity associated with fluid scale turbulence

- Energy gain from 2nd-order Fermi acceleration:

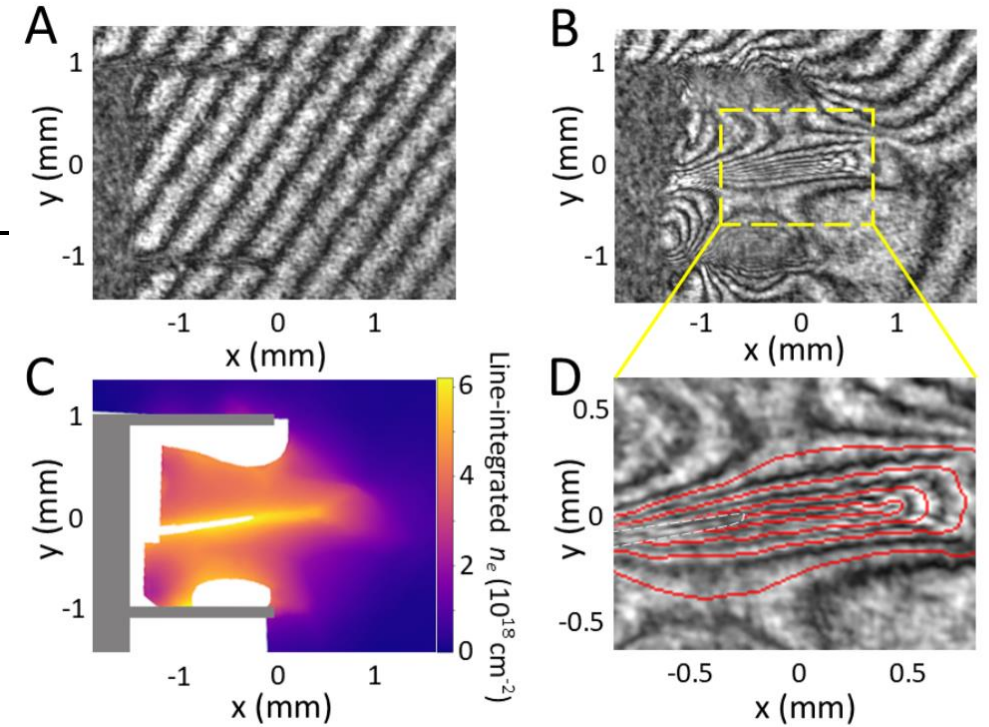
$$\Delta E \propto Z u_{\text{RMS}} B \sqrt{L_{\text{sys}} l_B}$$

- Residual density fluctuations are bounded by low-pass filtering each traced fringe at a cutoff set by the interaction-region width, taking the filtered trace as a baseline and subtracting it from the raw signal:

$$\frac{\delta n_e}{n_e} \lesssim \frac{\langle \delta(\int n_e \cdot d\ell) \rangle}{\langle \int n_e \cdot d\ell \rangle} \sqrt{\frac{L}{l_n}}$$

- Assuming continuity and subsonic, incompressible flow [Mohapatra & P. Sharma] density fluctuations can be associated with u_{RMS} :

$$u_{\text{RMS}} = c_s \sqrt{\frac{\delta n}{n}} \lesssim 38 \text{ kms}^{-1}$$



Mohapatra & P. Sharma “*Turbulence in the intracluster medium: simulations, observables, and thermodynamics*” Monthly Notices of the Royal Astronomical Society, 2019

Ion deflectometry provides bounds on the strength of the embedded magnetic fields

- Energy gain from 2nd-order Fermi acceleration:

$$\Delta E \propto Z u_{\text{RMS}} B \sqrt{L_{\text{sys}} l_B}$$

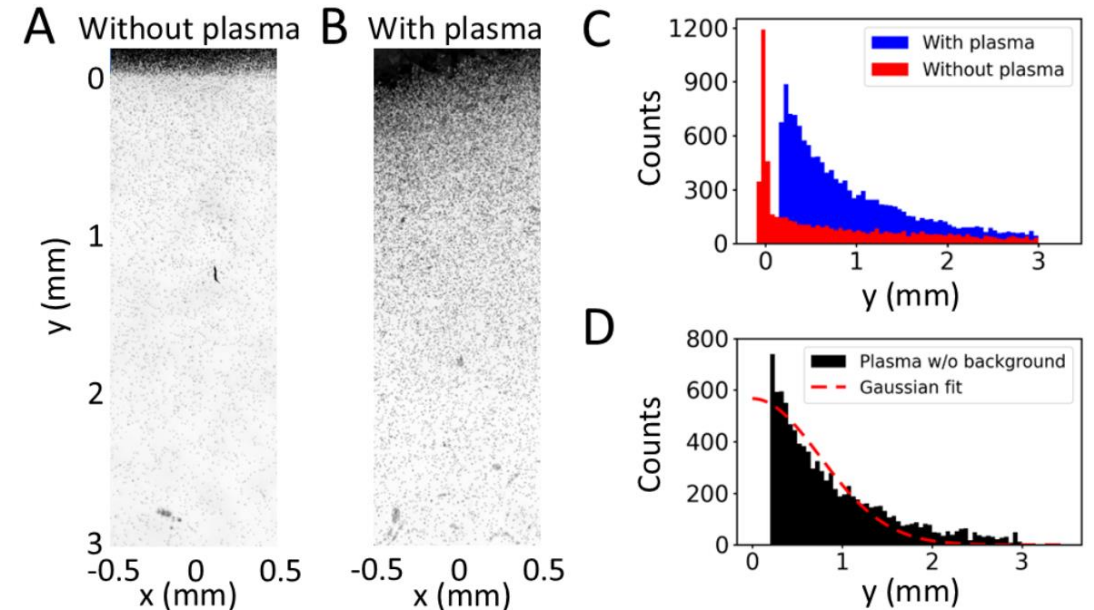
- A CR-39 detector placed in the ion beam path records ion tracks as pits

$$\langle \delta \theta \rangle \sim N \left(\frac{l_B}{r_c} \right)^2 \Rightarrow B_{\text{RMS}} \sim \frac{\langle \delta \theta \rangle}{Z_i B} \sqrt{\frac{2m_i E_i}{L_{\text{sys}} l_B}}$$

- Bound B_{RMS} depending on if the ions contributing to the signal arise from:

- Center of beam: $B_{\text{RMS}}^{\text{Max}} \lesssim 230 \text{ kG}$
- Position of knife edge: $B_{\text{RMS}}^{\text{Min}} \gtrsim 40 \text{ kG}$

- Advective electric-field contribution is $\lesssim 1\%$ (suppressed by $u/v_c \sim 10^{-2}$)



CR-39 integrates over ~ 1000 ion bunches. Only one coincides with the laser drive. **Compare deflection distributions with and without plasma to isolate the effect.**

Results indicate fluid-scale turbulence is too weak – observations are more consistent with a kinetic instability

- Combining the diagnostic constraints gives an upper bound on Fermi energisation:

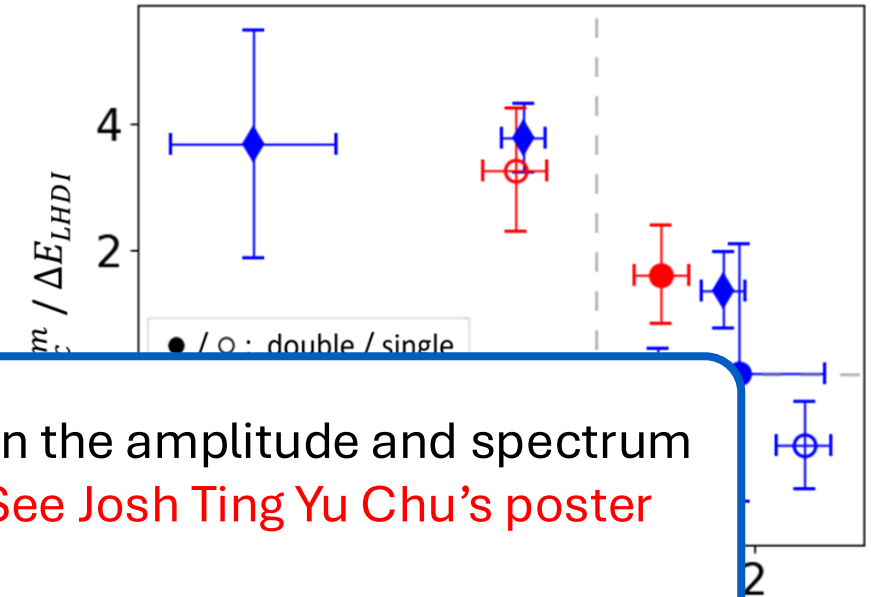
$$\Delta E_{\text{Fermi}} \sim Z_i e \frac{u_{\text{RMS}}}{c} B_{\text{RMS}} \sqrt{L l_B} \lesssim 13 \text{ keV}$$

- This is far below the observed energisation and broadening in velocity space

- A kinetic alternative: In future experiments, Thomson scattering can constrain the amplitude and spectrum of electrostatic fluctuations in the interaction region. **See Josh Ting Yu Chu's poster (P3.3051) this afternoon.**

- Expected energy change from LHDI is consistent with experiment:

$$\Delta E_{\text{LHDI}} \sim \kappa \frac{Z_i}{A_i n_e^{\frac{1}{4}}} B_{\text{RMS}} c_s \sqrt{L v_i} \sim 1 \text{ MeV}$$

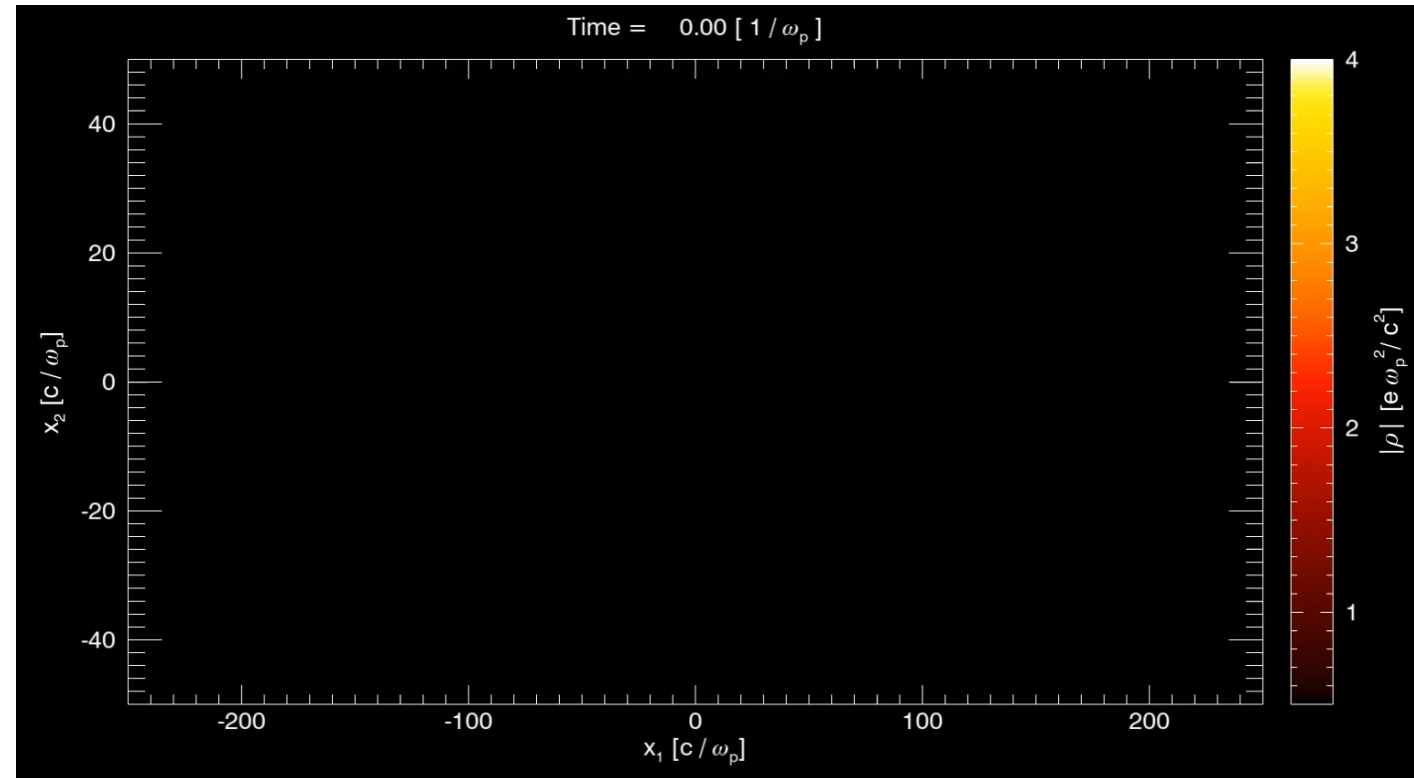


R. Bingham et al. “Particle acceleration by lower-hybrid turbulence” JPP. (2003).

T. Chang “Lower-hybrid collapse, caviton turbulence, and charged particle energization in the topside auroral ionosphere and magnetosphere” Phys. Fluids B. (1993).

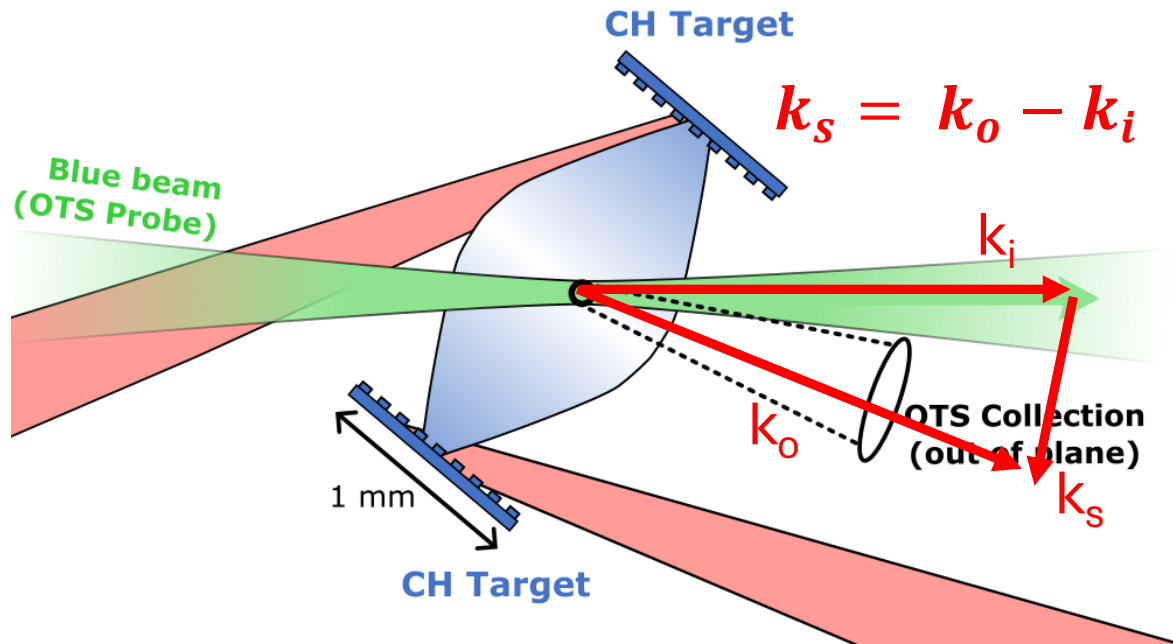
Reduced PIC simulations support the emergence of multi-scale drift physics

- Reduced 2D PIC setup isolates the basic interaction between a magnetized plasma flow and the experimental interaction region.
- Magnetized plasma flows in from $\pm y$ with periodic boundaries in x ; both ions and electrons are magnetized.
- Proton-to-electron mass ratio is 100 for numerical feasibility (qualitative results)
- Density structures develop on both ion and electron scales, suggestive of drift-type physics, potentially LHDI.



Courtesy of Pablo Bilbao (Oxford), performed with the OSIRIS Particle in Cell (PIC) code.

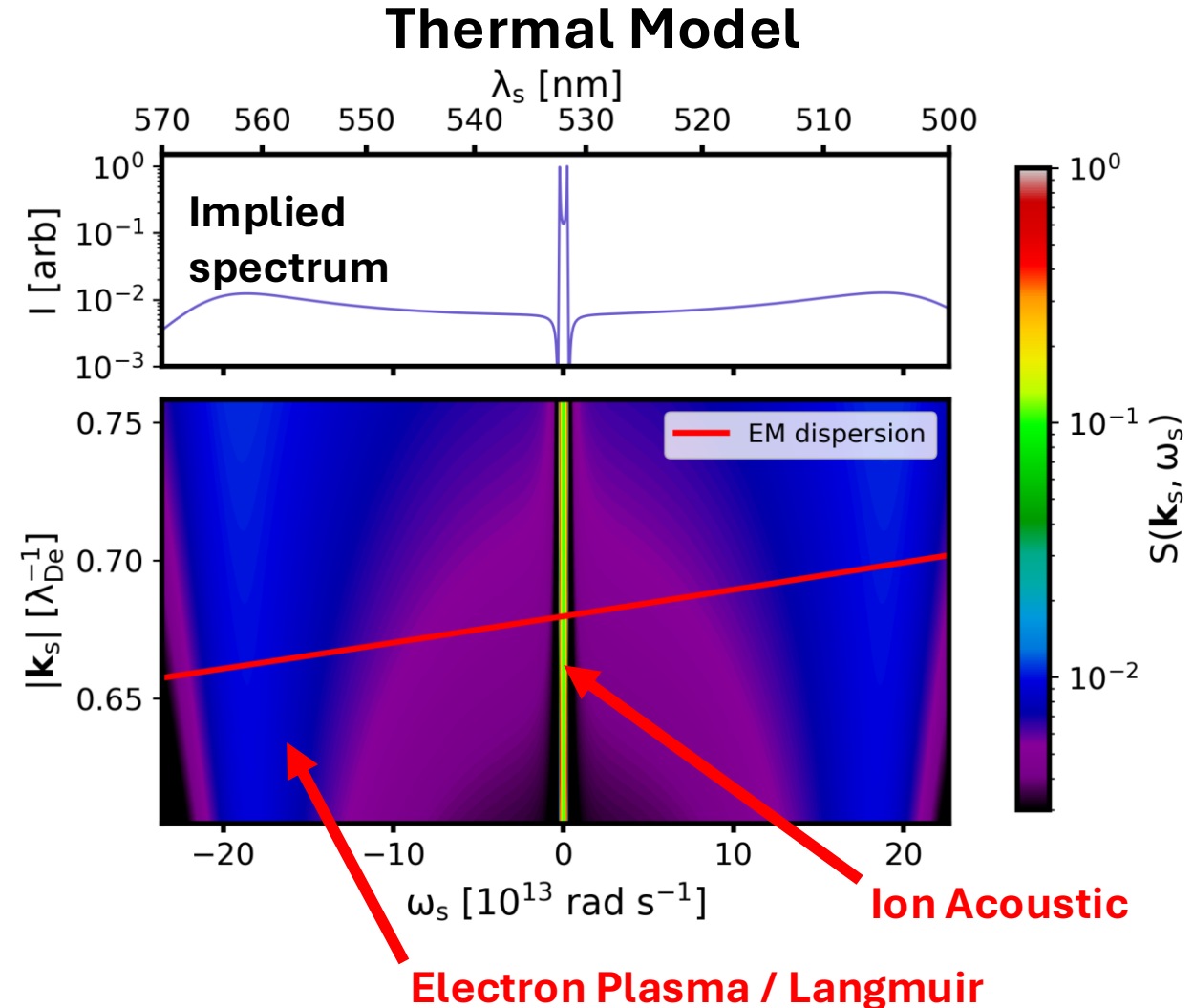
Optical Thomson Scattering (OTS) probes electrostatic fluctuations in localised volumes of plasma



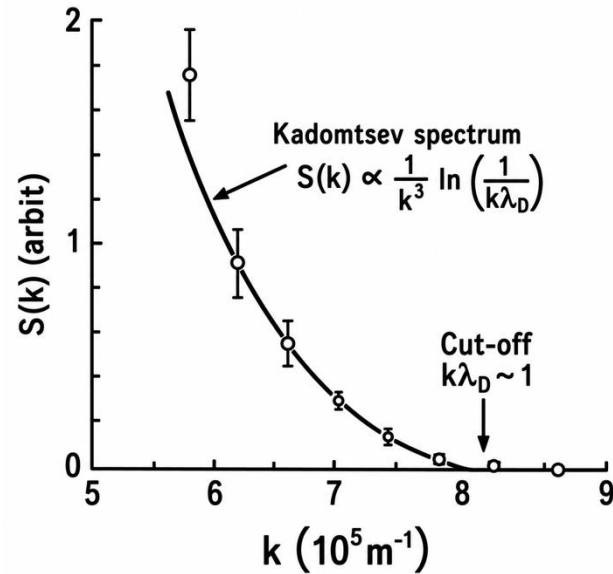
$$P_s(\lambda_s) \propto n_e \times S(\mathbf{k}_s, \omega_s)$$

Scattering geometry selects $\mathbf{k}_s$

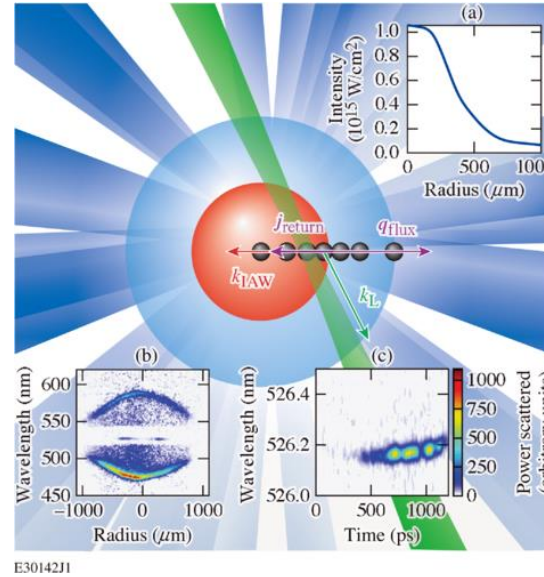
Spectral shift maps ω_s



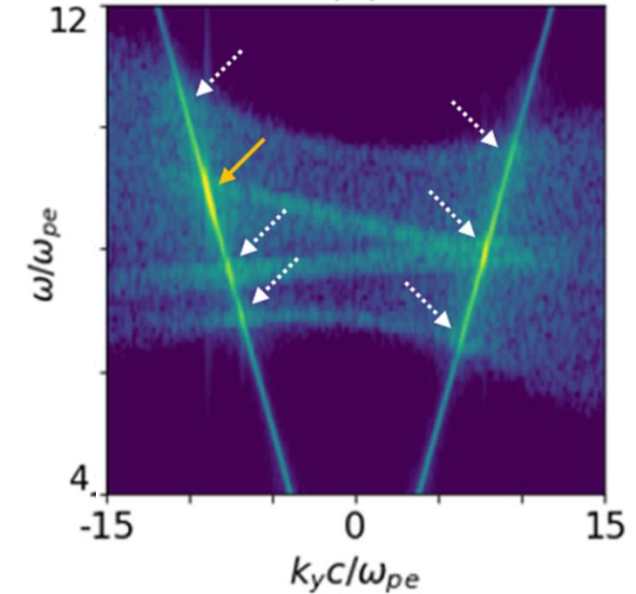
Electrostatic turbulence modifies the Thomson-scattering spectrum



Daughney et al. PRL (1970): scattering from a collisionless shock front. Measures the spectrum of current driven turbulence



Milder et al. PRL (2022): measured unstable IAW driven by a return-current instability to constrain models of non-local heat transport



Sato et al. PoP (2026): 2D PIC simulations show that beam-driven instabilities can amplify, distort the Thomson spectra

Takeaway: in future experiments, OTS can constrain the amplitude and spectrum of electrostatic fluctuations in the interaction region. **See Josh Ting Yu Chu's poster (P3.3051) this afternoon.**

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Laboratory observation of collective beam-plasma instabilities in a relativistic pair jet

Jack W. D Halliday^{1,2}, **Charlie D. Arrowsmith**^{2,3}, A. M. Goillot⁴, P. J. Bilbao⁵, P. Simon⁶, V. Stergiou^{2,4}, S. Zhang², P. Alexaki^{7,4}, M. Bochmann⁸, A. F. A. Bott², S. Burger⁴, H. Chen⁹, F. D. Cruz⁵, T. Davenne¹⁰, A. Dyson², A. Ebn Rahmoun⁴, I. Efthymiopoulos⁴, D. H. Froula³, J. T. Gudmundsson^{12,13}, D. Haberberger¹¹, T. Hodge¹⁴, S. Iaquinta², E. E. Los^{2,15}, G. Marshall¹⁴, F. Miniati², S. Parker¹⁵, B. Reville¹⁶, P. Rousiadou¹⁷, S. Sarkar², A. A. Schekochihin², K. G. Schlesinger⁸, L. O. Silva⁵, T. Silva⁵, R. Simpson⁹, E. Soria⁴, R. M. G. M. Trines¹, T. Vieu¹⁶, N. Charitonidis⁴, R. Bingham^{1,18} and **Gianluca Gregori**²

¹CLF, ²Oxford, ³LLE, ⁴CERN, ⁵IST, ⁶GSI, ⁷University of Athens, ⁸BoS / OuSoCo, ⁹LLNL, ¹⁰RAL, ¹²University of Iceland, ¹³KTH Royal Institute of Technology, ¹⁴AWE, ¹⁵Imperial College London, ¹⁶Max-Planck-Institut für Kernphysik, ¹⁷University of Ioannina, ¹⁸University of Strathclyde

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arXiv > physics > arXiv:2602.16542

Physics > Plasma Physics

[Submitted on 18 Feb 2026]

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Subjects: Plasma Physics (physics.plasm-ph)

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Article

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Laboratory realization of relativistic pair-plasma beams

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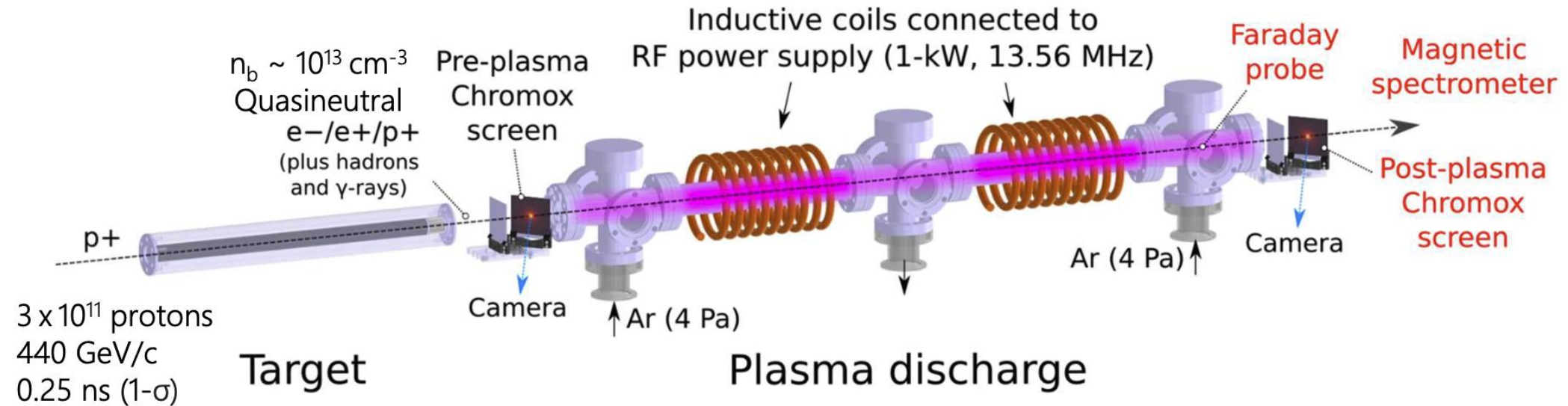
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C. D. Arrowsmith¹✉, P. Simon^{2,3}, P. J. Bilbao⁴, A. F. A. Bott¹, S. Burger², H. Chen⁵, F. D. Cruz⁴, T. Davenne⁶, I. Efthymiopoulos², D. H. Froula⁷, A. Goillot², J. T. Gudmundsson^{8,9}, D. Haberberger⁷, J. W. D. Halliday¹, T. Hodge^{1,10}, B. T. Huffman¹, S. Iaquinta¹, F. Miniati¹, B. Reville¹¹, S. Sarkar¹, A. A. Schekochihin¹, L. O. Silva⁴, R. Simpson⁵, V. Stergiou^{1,2,12}, R. M. G. M. Trines⁶, T. Vieu¹¹, N. Charitonidis², R. Bingham^{6,13} & G. Gregori¹

SPS proton beams generate relativistic electron positron jets for beam-plasma instability studies

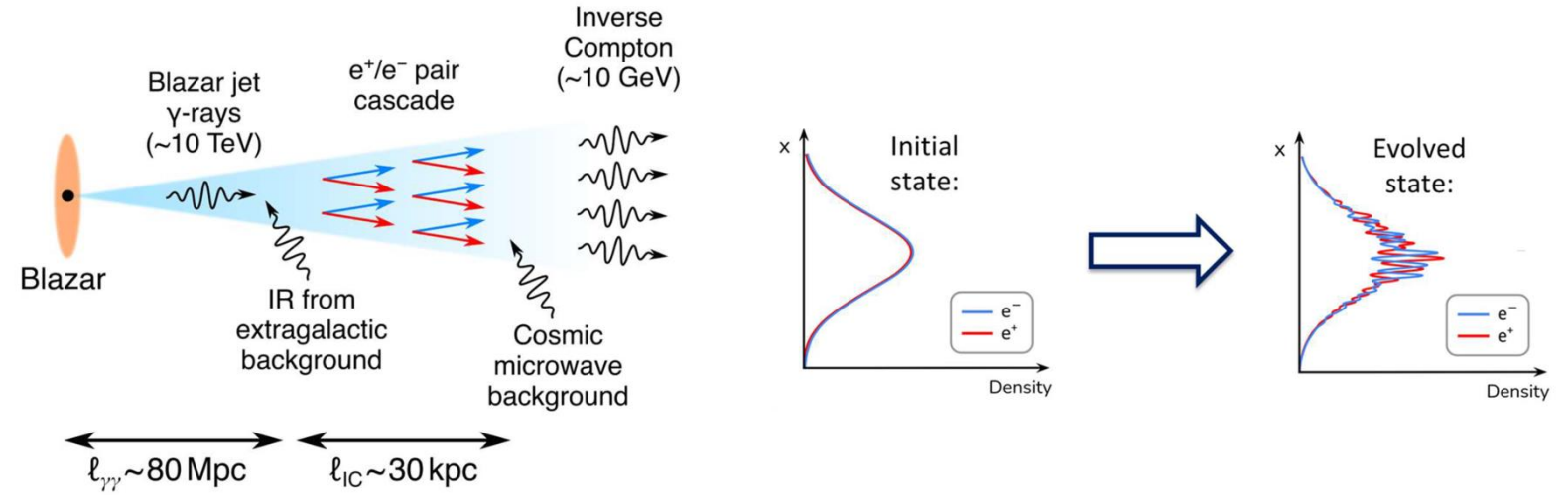


Beam density: $n_{\pm} \sim 10^{11} \text{ cm}^{-3}$
RF plasma density: $n_0 \sim 10^{12} \text{ cm}^{-3}$

} **Beam driven instability**

Motivation: Lab analogue of a Blazar Jet

Beam-plasma instabilities may shape blazar γ -ray spectra

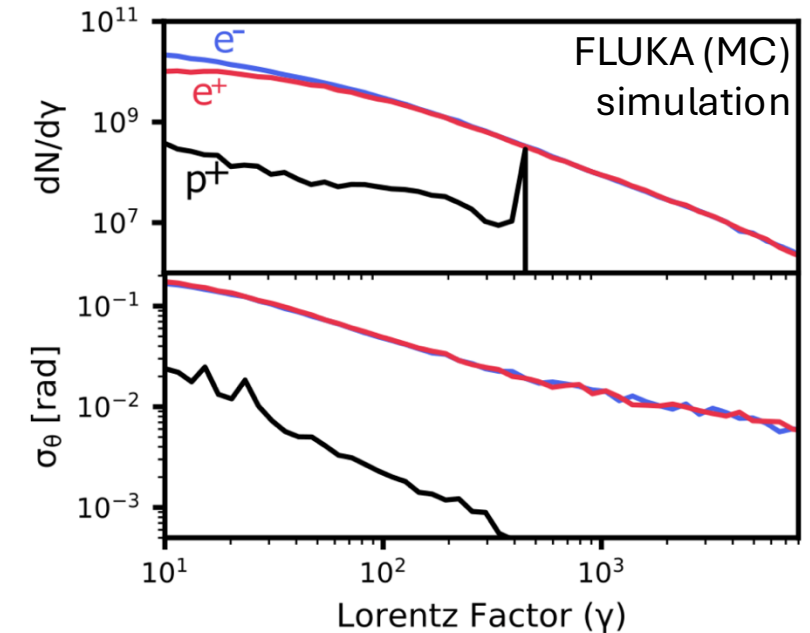
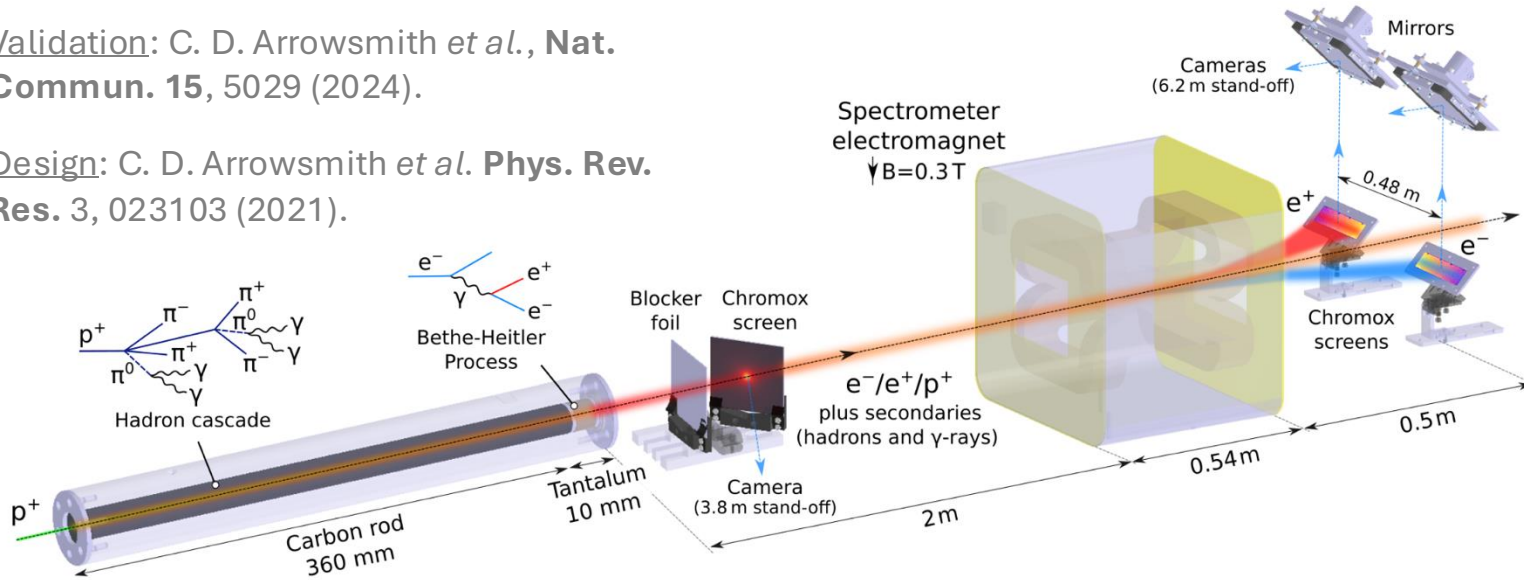


- Blazars are AGN jets pointed close to our line of sight, produce intense TeV γ -rays
- TeV photons generate relativistic pair beams in their interaction with extragalactic background light
- These pairs should inverse-Compton scatter CMB photons into the GeV γ -ray band
- Observed deficit of GeV emission is attributed to beam-plasma instabilities

Target-generated secondaries form broadband, non-thermal particle distributions

Validation: C. D. Arrowsmith *et al.*, **Nat. Commun.** **15**, 5029 (2024).

Design: C. D. Arrowsmith *et al.* **Phys. Rev. Res.** **3**, 023103 (2021).



Graphite target: Primaries initiate hadronic cascade, producing $\pi^\pm$, π^0 , photons and secondary hadrons.

Prompt π^0 decay: $\pi^0 \rightarrow \gamma\gamma$ produces GeV gamma-rays.

Tantalum converter: Gamma-rays drive an electromagnetic cascade, producing relativistic $e^\pm$ pairs

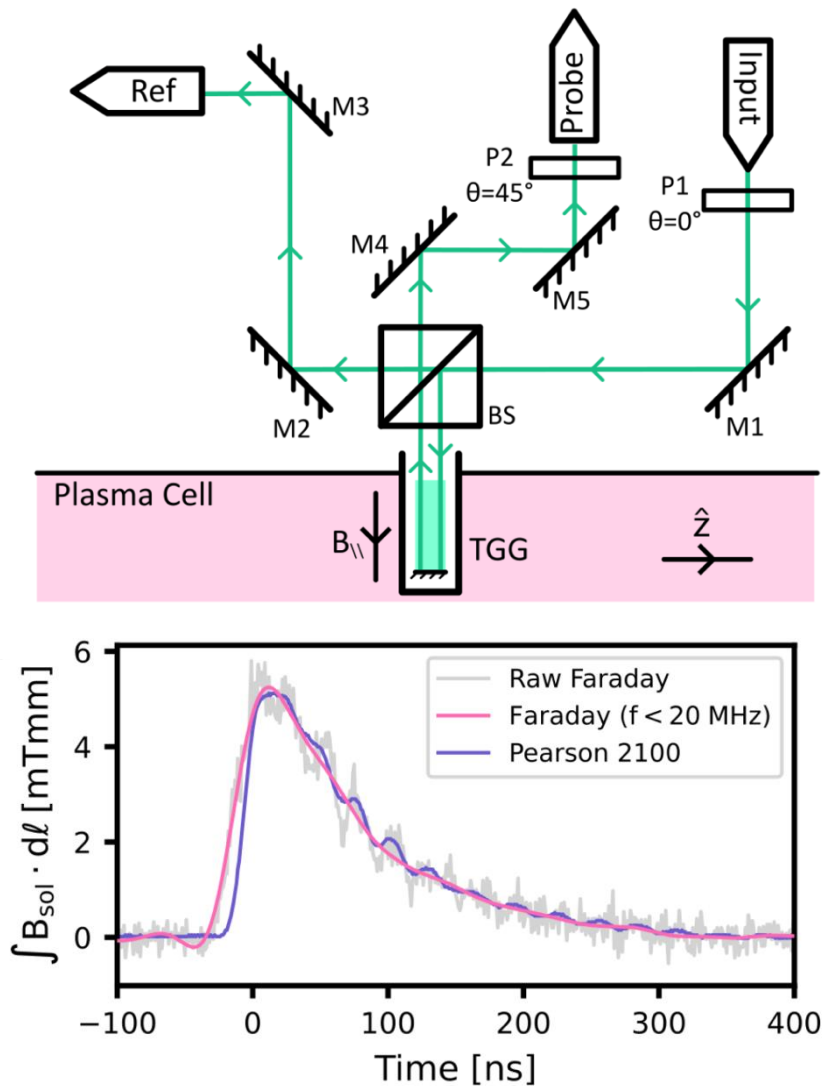
Takeaway: The target converts clean SPS protons into a broadband, relativistic, pair-rich population resembling an astrophysical non-thermal particle distribution.

A Faraday-rotation probe was used to constrain path-integrated magnetic fields

- Linearly polarised laser directed along the axis of a magneto-optic terbium gallium garnet (TGG) crystal.
- Via the Faraday effect, the polarisation is rotated through α :

$$\alpha = \mathcal{V} \int \mathbf{B} \cdot d\mathbf{l}$$

- For our instrument, $\mathcal{V} \sim 220 \text{ rad T}^{-1} \text{ m}^{-1}$
- Rotation mapped to an intensity change via Malus' law
- Measurement is line-integrated: constrains spatially averaged field not local structure
- Time response and sensitivity characterised offline.

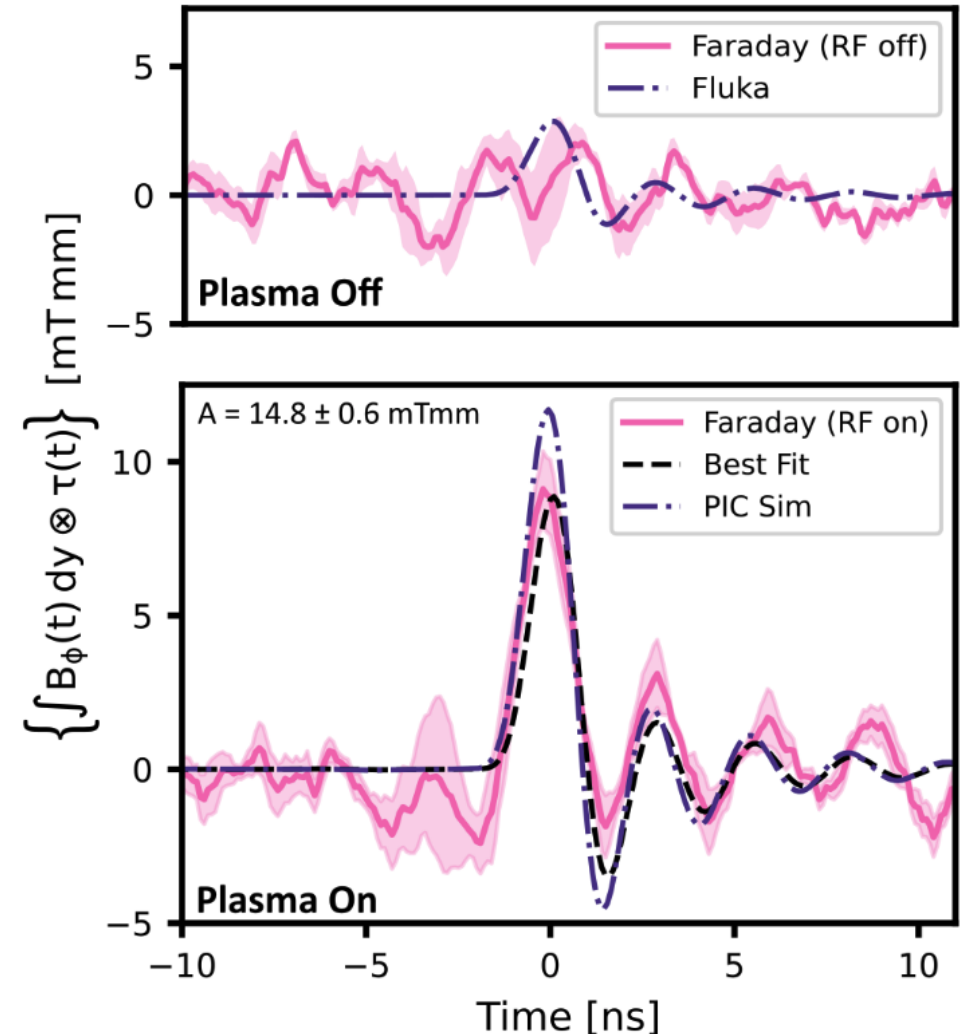


The presence of an ambient RF discharge plasma was observed to amplify magnetic fields advected with the jet

RF plasma on: coherent signal of ~ 15 mT mm observed in the path-integrated field.

RF plasma off: no comparable magnetic-field signal is detected.

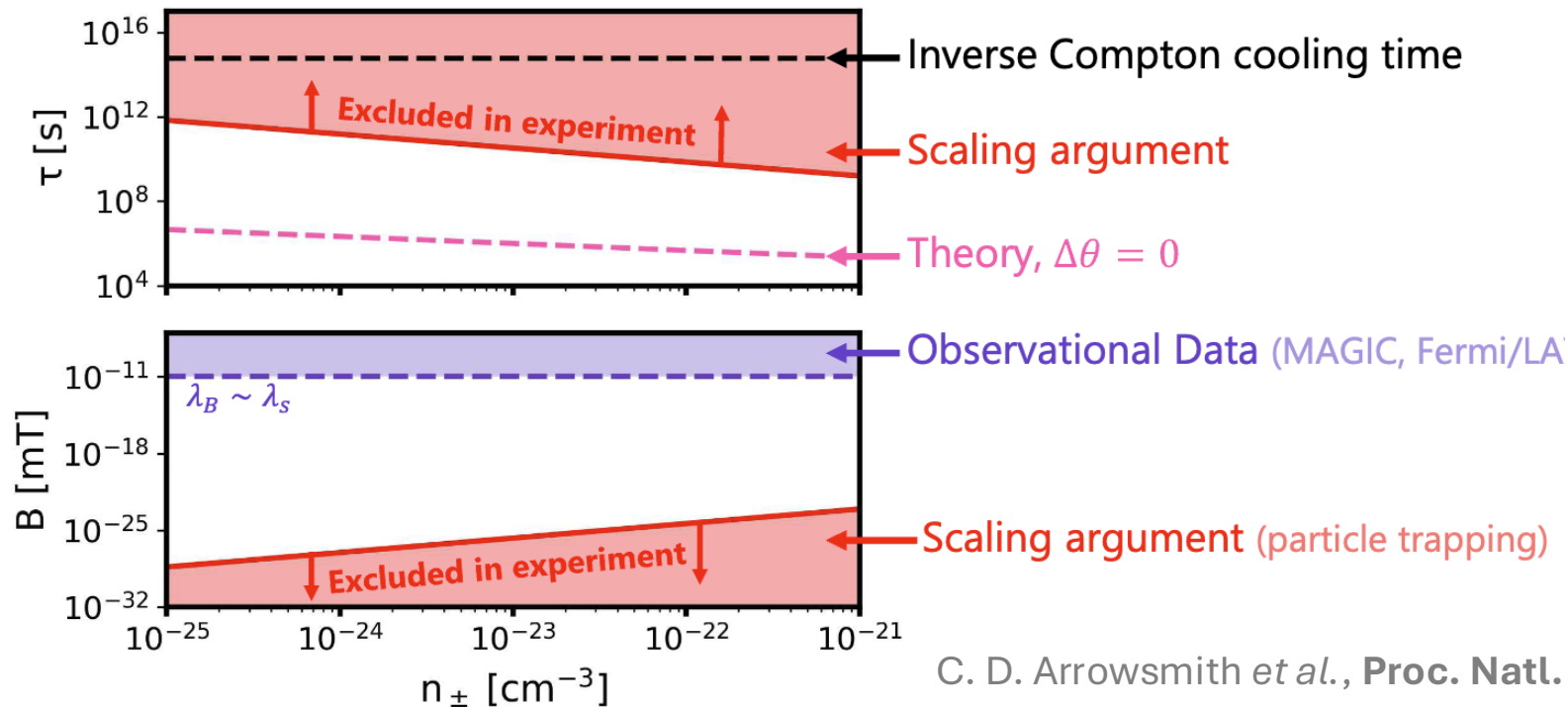
- Traces show the average of 3 shots; shaded region gives shot-to-shot standard deviation.
- Probe-light-off backgrounds are subtracted to remove radiation-induced noise.
- PIC simulations reproduce the signal when residual protons and pair-beam divergence are included.



Simulations: C. D. Arrowsmith *et al.*, **Proc. Natl. Acad. Sci.** 122 (45) e2513365122 (2025)

Measurement: J. W. D Halliday *et al.* **arXiv:2602.16542**

Measured beam-plasma growth rates inform blazar-cascade models



Theoretical growth rate:

$$\Gamma_{\text{fastest}} = \sqrt{2/3} \frac{\omega_p}{\Delta\theta} \left(\frac{n_{\pm}}{2n_p \gamma_{\pm}} \right)^{\frac{2}{3}} \quad \left\{ \Delta\theta \gg \left(\frac{n_{pm}}{2n_p \gamma_{\pm}} \right)^{1/3} \right\}$$

Scaling relation:

$$\Gamma_{\text{blz,sc}} [\text{s}^{-1}] \leq 3 \times 10^{-11} \left(\frac{\Gamma_{\text{exp}}}{0.7 \text{ ns}^{-1}} \right)$$

C. D. Arrowsmith *et al.*, **Proc. Natl. Acad. Sci.** 122 (45) e2513365122 (2025)



O5.26 Eva Los and O5.28 Celia Stergiou – Friday afternoon, Tinto

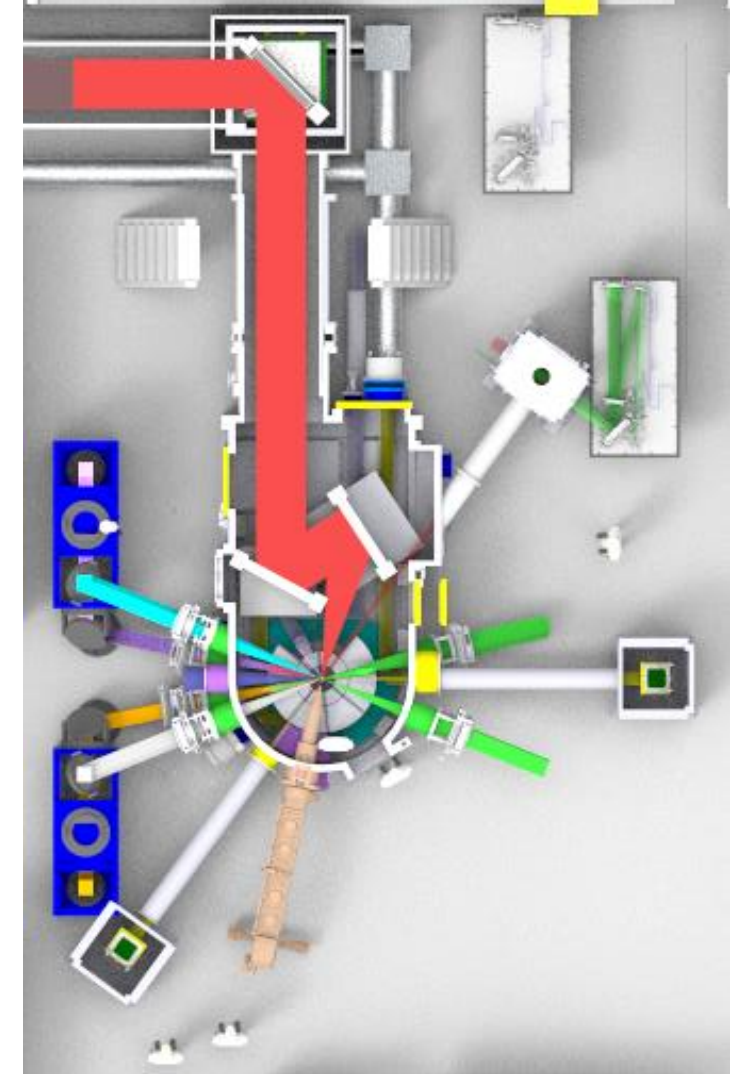
P3.3054 Max Mudra – This afternoon (BSAP03, poster)

Overview

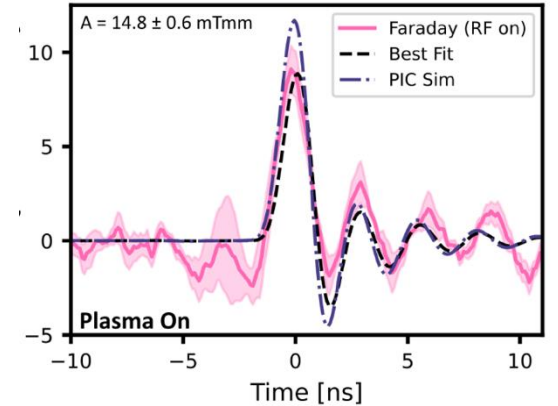
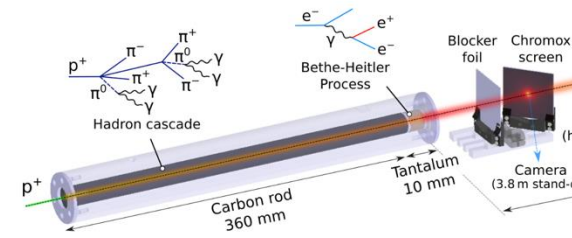
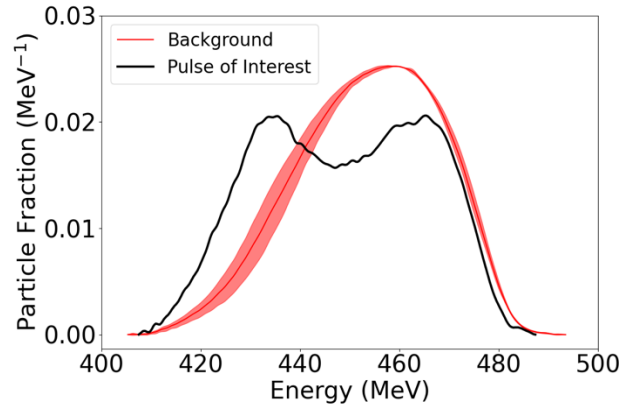
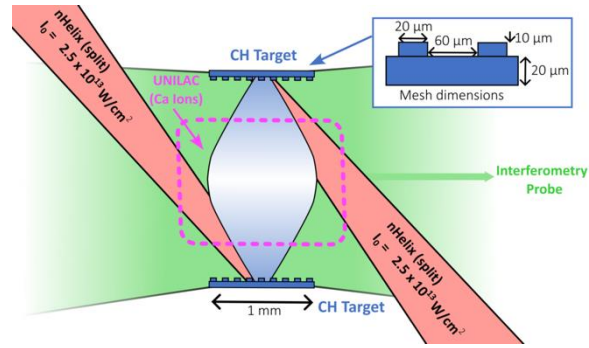
- **Astrophysical problem** – How do collisionless plasmas exchange energy between bulk flows, electromagnetic fluctuations, and non-thermal particles?
- **Laboratory strategy:** use accelerator beams as controllable fast-particle populations
- **Case study 1:** Ion acceleration / diffusion by laser driven, magnetized plasma:
 - Experimental design
 - Time of Flight / interferometry
- **Case study 2:** Relativistic pair jets interacting with an ambient RF plasma:
 - Experimental design
 - Faraday rotation diagnostic
 - Application to Blazar jets
- **Outlook:** next-generation laser facilities can turn these into systematic, higher-repetition, diagnostic-rich studies.

Next generation laser facilities can address some related physics on a more systematic basis

- **Vulcan 20-20:** combines kJ, 10 ns-class laser beams with 1 PW and 20 PW short-pulse capability.
- **Long-pulse beams:** create thermal plasmas with astrophysically relevant bulk properties, including fluid-scale turbulence.
- **Short-pulse beams:** inject fast ion and electron populations, for example via TNSA and LWFA.
- **Physics reach:** collisionless shock formation, energetic-particle injection, acceleration and scattering.
- **Key advantage:** higher access and richer diagnostic capability than conventional accelerator platforms.



Conclusions



- Particle accelerators provide controlled, fast-particle beams that can be used to model high-energy astrophysics terrestrially
- GSI Ion beam data showed evidence of acceleration / diffusion not explained by fluid-scale processes, pointing towards kinetic instability
- CERN pair-jet experiments accessed conditions relevant to relativistic jets
- Beam-plasma systems will be accessible on next generation lasers including Vulcan 20-20